♡ cast of characters ♡

capacitors
more capacitance, more chonk
(the wires go in one end and out the other)

10pF 1nF 100nF 1µF 4µ7F 22µF

resistor
grumpy, throws bricks
(and the odd molotov)

ferrite bead
a very hungry inductor

crystal
dances at its own frequency ♪

FPGA
Stimulus & Sampling Pulse Timing Generation

Pulse Amplifier

Signal Splitter
Couples incident stimulus into mesh, reflections into sampler

L/R Connector Mux

Termination Mux & Crossbar
Selects termination, and flips mesh send/return

Sub-Mesh Output Mux

Sampling Bridges

Clip Lines

Low-Speed Amplifiers

Microcontroller

Debug USB

Debug connector

PiHAT+ Connector

Power Supplies
5V input, outputs: 3V3D (digital), 3V3A (analog, low noise), 1V8 (digital), 1V2 (digital)

let's sense some pillows~! ♡

✦ センサー回路図 ✦

yanartas-pillowbox-sensors

~ the pillowbox sensor board ~

sheet ♡ 1 / 1

KiCad 10 · paper A1 · drawn with love ♪